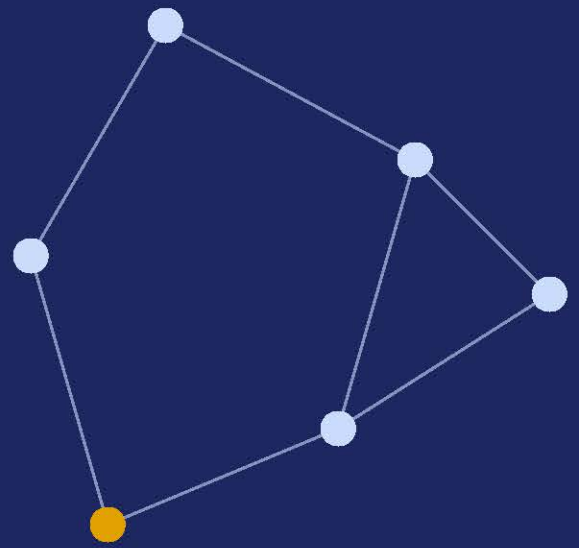
ARTIFICIAL INTELLIGENCE



Lecture 1 — Introduction

From "what is intelligence" to the first formal notion of an agent — now with worked examples

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Today's Agenda

1. What is Intelligence?
2. What is Artificial Intelligence?
3. Four Main Approaches to AI
4. The Turing Test & Total Turing Test
5. The Chinese Room Argument
6. Strong vs. Weak AI
7. Where Are We Today?
8. Intelligent Agents

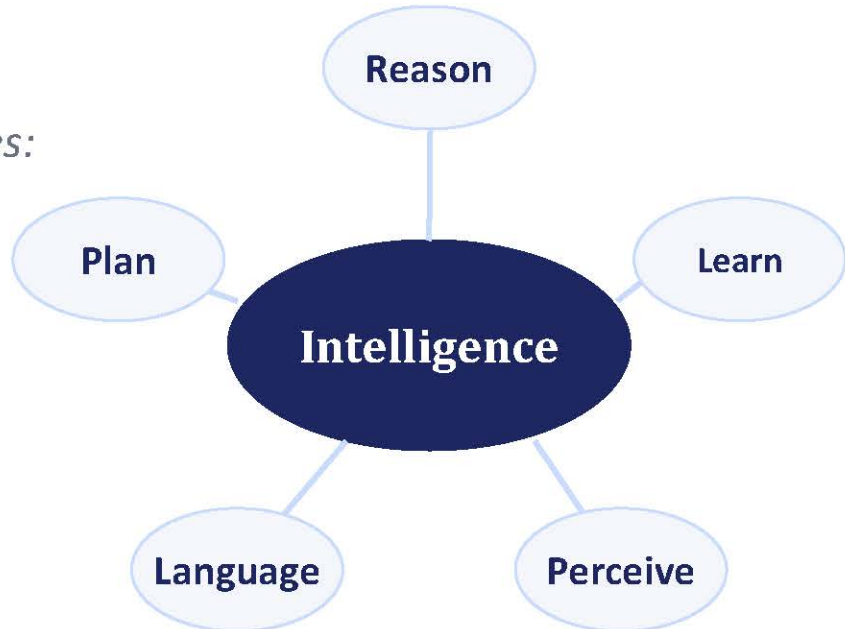
Every concept is followed by a worked example or comparison table.

What is Intelligence?

A working definition: the ability to acquire and apply knowledge, reason about a situation, adapt to new circumstances, and solve problems in pursuit of a goal.

Psychology and cognitive science point to several component abilities:

- Reasoning
- Learning
- Perception
- Language
- Planning



EXAMPLES & COMPARISON

Intelligence in Practice

Ability	Human Example	Animal Example	Machine Example
Reasoning	Solving a logic puzzle	A crow bending wire to reach food	Expert system diagnosing a fault
Learning	Learning a new language	A dog learning tricks by repetition	Neural network trained on labeled images
Perception	Recognizing a face in a crowd	A bat using echolocation	Computer vision detecting objects
Language	Holding a conversation	A dolphin communicating via clicks	A chatbot answering customer queries
Planning	Planning a road trip	Ants finding the shortest path to food	A robot planning a collision-free path

Takeaway: the same underlying ability shows up differently across humans, animals, and machines — AI tries to reproduce the ability, not necessarily the mechanism.

What is Artificial Intelligence?

The design and study of computational agents that perceive their environment and act to achieve goals — i.e., building systems that exhibit the traits of intelligence in a machine.

1

Vision

Interpreting images and video (object detection, medical imaging)

2

Language

Understanding and generating text and speech

3

Robotics

Sensing and acting in the physical world

4

Decision-Making

Planning, search, and game-playing under constraints

EXAMPLES & COMPARISON

AI You Already Use

Domain	Everyday Example	Underlying Technique
Vision	Face ID unlocking your phone	Convolutional neural networks
Language	Voice assistants (Siri, Alexa)	Natural language processing / LLMs
Robotics	A robotic vacuum navigating a room	Sensor fusion + motion planning
Decision-Making	Netflix recommending a show	Machine learning / optimization

Takeaway: AI is not one algorithm — it's a collection of techniques, each matched to a different kind of perception, reasoning, or action problem.

Four Main Approaches to AI

Defined along two axes: what we imitate (human vs. ideal rationality) and what we model (thought vs. behavior).

	HUMANLY	RATIONALLY
THINKING	**Thinking Humanly** Cognitive modeling: mimic the reasoning steps of the human mind. *e.g., cognitive architectures (SOAR, ACT-R)*	**Thinking Rationally** Laws of thought: formal logic to derive correct conclusions. *e.g., logic-based inference engines*
ACTING	**Acting Humanly** Behave indistinguishably from a human (Turing Test). *e.g., chatbots, natural-language interfaces*	**Acting Rationally** Rational agent: act to achieve the best expected outcome. *→ the approach this course follows*

EXAMPLES & COMPARISON

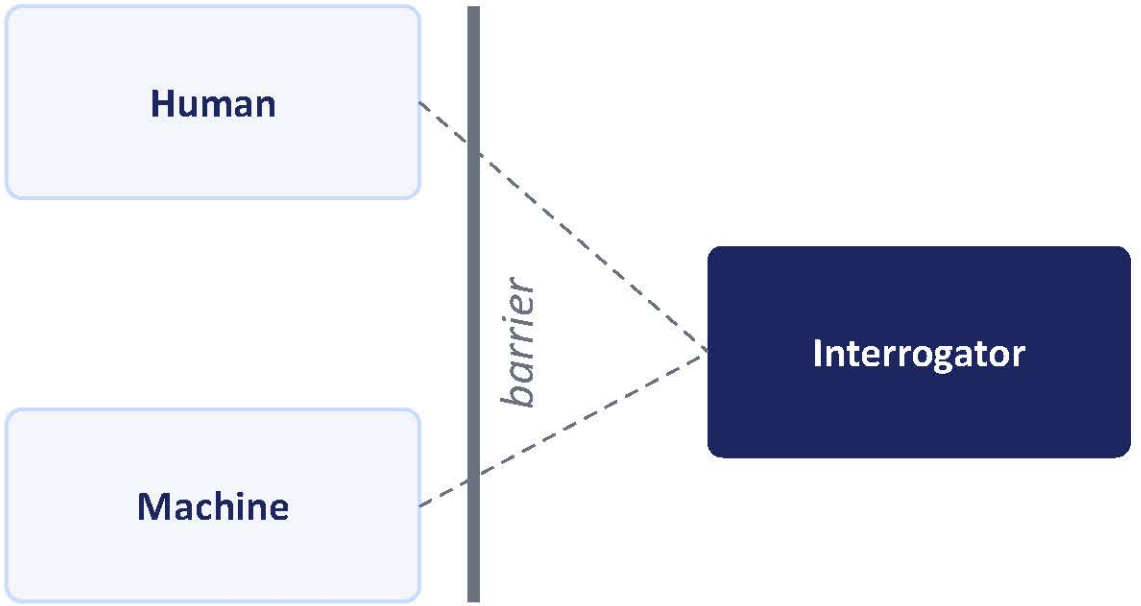
Approaches Compared

Approach	Example System	Strength	Weakness
Thinking Humanly	SOAR / ACT-R cognitive models	Insight into human cognition	Hard to verify against real thought
Thinking Rationally	Prolog-based logic solvers	Provably correct inference	Struggles with uncertainty & scale
Acting Humanly	ELIZA-style chatbots	Naturally engages users	Can fake understanding — no real reasoning
Acting Rationally	Self-driving cars, game-playing agents	Optimizes for a measurable goal	Only as good as the goal you define

Takeaway: this course builds systems the "acting rationally" way — defining a performance measure and designing an agent that optimizes it.

The Turing Test (1950)

Alan Turing proposed the “imitation game”: an interrogator holds a text-only conversation with a hidden human and a hidden machine, and must decide which is which. If the interrogator cannot reliably tell them apart, the machine is said to have passed.



Why it's a milestone:

- Avoids defining “thinking” directly — tests behavior instead
- Operational, measurable criterion for machine intelligence
- Criticized: rewards mimicry/deception, ignores understanding

EXAMPLES & COMPARISON

Famous Attempts, Through History

Attempt	Year	Result
ELIZA	1966	Fooled some users with simple pattern matching — no real understanding
PARRY	1972	Simulated a paranoid patient; fooled psychiatrists in blind trials
Loebner Prize entries	1991–present	Annual contest; no entrant has convincingly passed the unrestricted test
Modern LLMs (e.g., ChatGPT)	2022+	Often mistaken for human in short, informal exchanges

Takeaway: passing informally is easy; passing a rigorous, extended, expert-led test remains an open bar nobody has clearly cleared.

The Total Turing Test

Extends Turing's original test by requiring the machine to demonstrate intelligence through physical interaction with the world — not just conversation.

Turing Test

- Text-only conversation
- Tests language & reasoning
- No perception or action required

Total Turing Test

- Adds a video/perceptual channel
- Adds physical manipulation of objects
- Requires: vision, robotics, plus language & reasoning

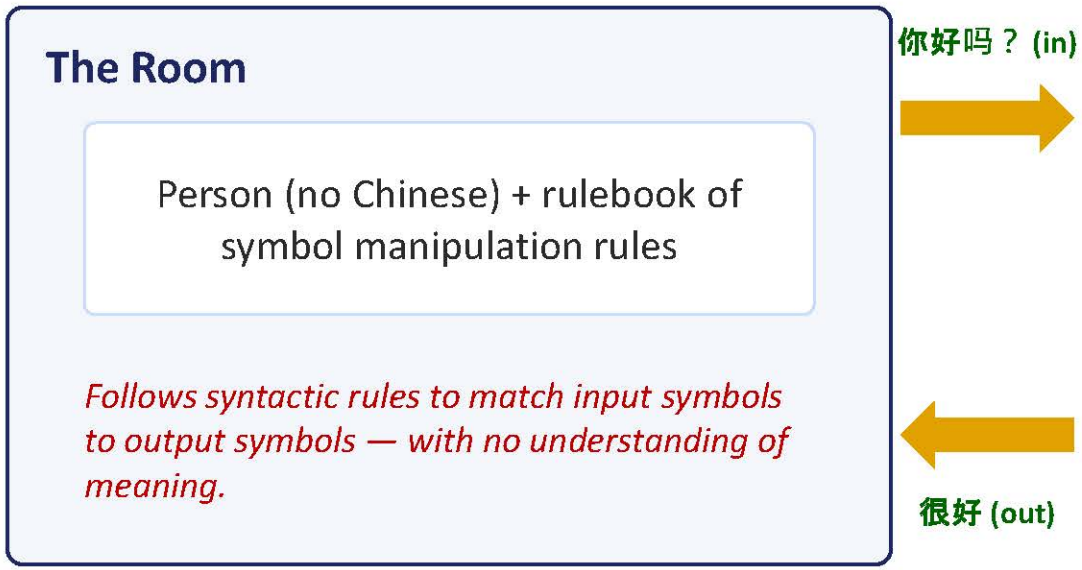
EXAMPLES & COMPARISON

Who Would Actually Pass?

System	Perceives & Acts in the World?	Converses in Language?	Passes the Total Turing Test?
Text-only chatbot	No	Yes	No
Autonomous mobile robot	Yes	No	No
Research humanoid (vision + manipulation + dialogue)	Yes	Partial	Closest so far
A human	Yes	Yes	Yes (by definition)

Takeaway: adding perception and action raises the bar substantially — no current system combines all three convincingly.

The Chinese Room Argument (Searle, 1980)



Searle's claim: syntax (symbol manipulation) is not sufficient for semantics (genuine understanding) — so passing the Turing Test does not imply a mind.

Common replies:

- Systems Reply — the person alone doesn't understand, but the whole system (person + rules + room) might
- Robot Reply — grounding symbols in sensors/actuators could produce real understanding
- Other Minds Reply — we infer human understanding from behavior too; why not a machine's?

EXAMPLES & COMPARISON

Applying the Argument to Modern AI

Position	Core Claim	Applied to a Modern LLM
Searle (original)	Syntax manipulation ≠ semantics/understanding	It predicts plausible next tokens, but arguably doesn't "understand" meaning
Systems Reply	Understanding emerges from the whole system, not the person alone	The full model + training + weights might understand, even if no single part does
Robot Reply	Grounding in sensors/actuators enables real understanding	An embodied LLM connected to a robot might understand more meaningfully
Other Minds Reply	We infer human understanding from behavior — why not machines?	If its behavior is indistinguishable from a human's, on what basis do we deny understanding?

Takeaway: this debate is not settled — and it is exactly the question people ask today about large language models.

Strong AI vs. Weak AI

Weak AI

AI as a tool

- Simulates intelligent behavior
- No claim of genuine mind or consciousness
- Goal: useful, well-performing systems
- Examples: search engines, recommender systems, game AI

Strong AI

AI as a mind

- Claims genuine understanding & consciousness
- The program IS a mind, not just a model of one
- Philosophically contested (see: Chinese Room)
- Examples: hypothetical Artificial General Intelligence (AGI)

EXAMPLES & COMPARISON

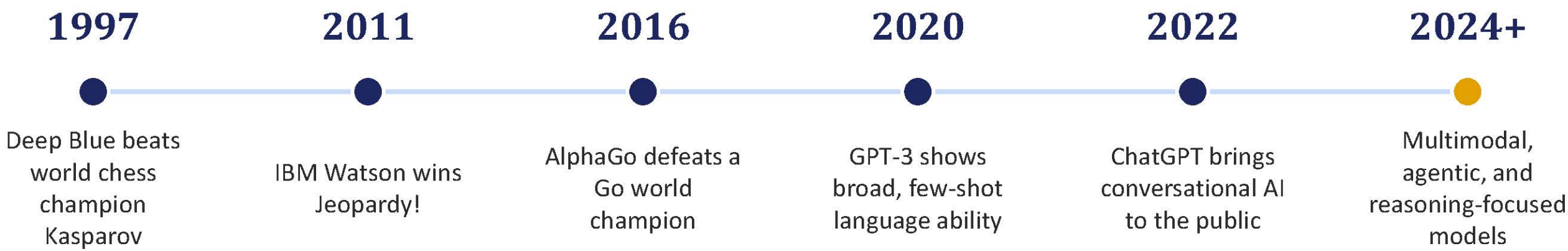
Classifying Real Systems

System	Weak or Strong?	Why
Siri / Alexa	Weak	Simulates conversation; no genuine understanding is claimed
Deep Blue (chess)	Weak	Excellent at one narrow task; no general mind
Watson (Jeopardy!)	Weak	Broad but still task-specific pattern matching
Hypothetical AGI	Strong (theoretical)	Would possess genuine understanding across domains
The human brain	Strong (by definition)	The only confirmed example of general intelligence so far

Takeaway: nearly everything deployed today is Weak AI — Strong AI remains a philosophical goalpost, not a shipped product.

Where Are We Today?

Milestones that reframed what people believed AI could do:



Takeaway: progress has moved from narrow, rule-based expertise toward broad, learned, interactive capability — setting up the rest of this course.

EXAMPLES & COMPARISON

Four Eras, Compared

Era	Representative System	Key Technique	Limitation
1980s: Expert Systems	MYCIN (medical diagnosis)	Hand-coded rules	Brittle outside its rules
1990s–2000s: Classical ML	Spam filters, credit scoring	Statistical learning (SVMs, trees)	Needs hand-engineered features
2010s: Deep Learning	ImageNet-winning CNNs, AlphaGo	Deep neural networks	Data-hungry, hard to interpret
2020s: Large Language Models	ChatGPT and similar models	Transformers on massive data	Can be confidently wrong (“hallucinates”)

Takeaway: each era didn't replace the last so much as add a new capability layer — rules, statistics, representation learning, and now scale.

Intelligent Agents: A First Look

An agent is anything that perceives its environment through sensors and acts upon that environment through actuators. Its behavior is described by an agent function mapping percept histories to actions.

Coming up in Lecture 2:

- PEAS: Performance, Environment, Actuators, Sensors
- Agent types: reflex, model-based, goal-based, utility-based, learning
- Environment properties: observable, deterministic, episodic, static, discrete



EXAMPLES & COMPARISON

Everyday Systems as Agents

System	Percepts	Actions	Agent Type
Thermostat	Room temperature	Turn heater on/off	Simple reflex
Robotic vacuum	Bump & dirt sensors	Move, turn, vacuum	Model-based reflex
GPS navigation app	Location, traffic data	Suggest / re-route path	Goal-based
Self-driving car	Cameras, lidar, GPS	Steer, brake, accelerate	Utility-based
Recommendation system	Clicks, ratings, history	Suggest items	Learning agent

Takeaway: the agent categories aren't just theory — you interact with all five types before lunchtime most days.

Key Takeaways

- ✓ Intelligence is a bundle of abilities: reasoning, learning, perception, language, planning.
- ✓ AI can be defined along two axes — thinking vs. acting, human-like vs. rational.
- ✓ This course adopts the acting-rationally (rational agent) view of AI.
- ✓ The Turing Test measures behavior; the Total Turing Test adds perception & action.
- ✓ The Chinese Room argues that symbol manipulation alone may not be understanding.
- ✓ Weak AI builds useful tools; Strong AI claims a genuine mind — the debate remains open.
- ✓ An agent perceives via sensors and acts via actuators — our unit of analysis going forward.

Next: Lecture 2

Intelligent Agents

PEAS descriptions · agent architectures · environment properties · a hands-on agent lab